

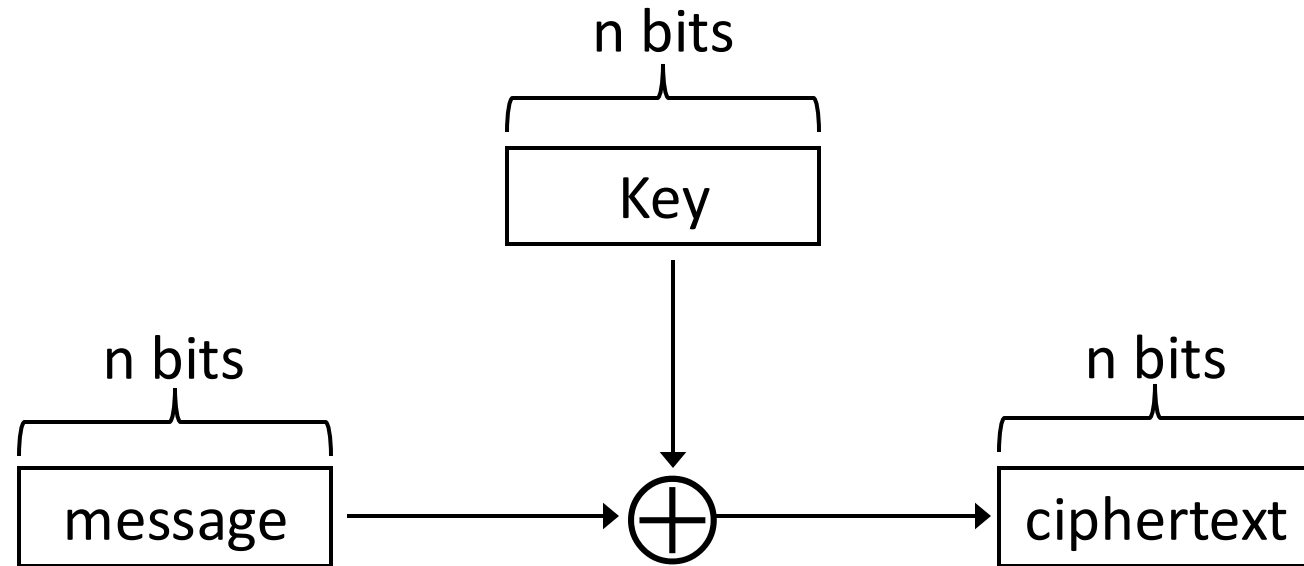
Stream Cipher and PRG

The bad news ...

Thm: perfect secrecy $\Rightarrow |\mathcal{K}| \geq |\mathcal{M}|$

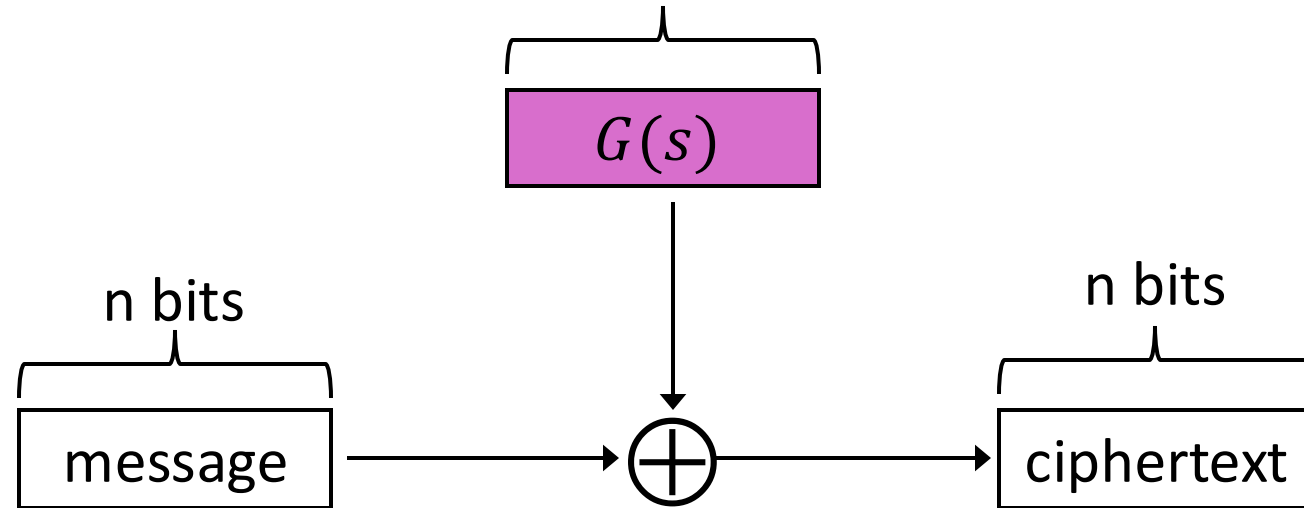
Idea: replace “random” key by “pseudorandom” key

One-time pad



Stream Cipher

$$G: \{0, 1\}^s \rightarrow \{0, 1\}^n$$



- Efficiently computable by a deterministic algorithm
- $s \ll n$

Can a stream cipher have perfect secrecy?

- A. Yes, if the PRG is really “secure”
- B. No, there are no ciphers with perfect secrecy
- C. Yes, every cipher has perfect secrecy
- D. No, since the key is shorter than the message

Not possible!!!

- Size of key space has to be at least the same as the message space $|\mathcal{K}| \geq |\mathcal{M}|$ to achieve perfect secrecy

Computational security

Two relaxations:

1. Security only against **efficient** adversaries.
2. Security can fail with a **very small probability**

Relaxation 1: efficient adversary

Polynomial time: $O(n^c)$ for any constant c

$O(n^3)$ vs. $O(2^n)$

2^{60} seconds since the start of the universe

Security against all probabilistic polynomial time (PPT) adversaries

Relaxation 2: small failure probability

Negligible function f : for every polynomial p , there is an integer N such that for all $n > N$,

$$f(n) < \frac{1}{p(n)}$$

E.g.,: $2^{-n} = \frac{1}{2^n}$, $2^{-\sqrt{n}}$, $n^{-\log n}$

How small is 2^{-60} ?

2^{-14}



[NEWSNATION]

According to the NWS, a person has a 1-in-15,300 chance of getting struck by lightning in their lifetime.

It's said you are four times more likely to get a perfect score on your SAT than getting struck by lightning.

Negligible function $\text{negl}(n)$

$\text{negl}_1(n) + \text{negl}_2(n)$ is negligible

$\text{negl}_1(n) \cdot p(n)$ is negligible for any polynomial p

Definition of computational security

- Algorithm $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$:
 - $k \in \mathcal{K}$ secret key, $m \in \mathcal{M}$: message, $c \in \mathcal{C}$: ciphertext
 - $k = \text{Gen}(\mathbf{1}^n)$
 - $c = \text{Enc}_k(m)$
 - $m = \text{Dec}_k(c)$
- Correctness: $m = \text{Dec}_k(\text{Enc}_k(m))$

n : security parameter
usually use λ

Computation indistinguishability against eavesdropper

Experiment $\text{PrivK}_{A,\Pi}^{\text{eav}}(n)$:

1. Adversary A is given input 1^n , and outputs two messages $m_0, m_1 \in \mathcal{M}$
s.t. $|m_0| = |m_1|$
2. A key k is generated using $\text{Gen}(1^n)$, a random bit $b \in \{0,1\}$ is chosen. $c = \text{Enc}_k(m_b)$ is given to A .
3. A outputs b'
4. Experiment outputs 1 if $b=b'$, and 0 otherwise.

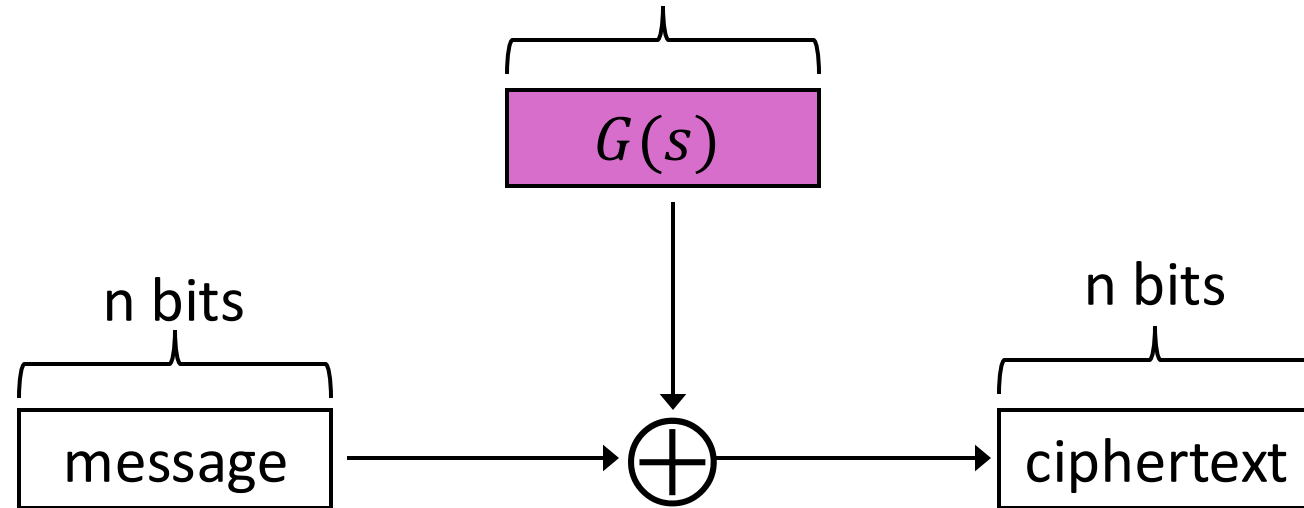
Computational Indistinguishability against eavesdropper

Definition (EAV security): a private-key encryption $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ is **EAV-secure**, if for all **probabilistic polynomial time** adversaries A , there is a **negligible function** negl such that for all n ,

$$\Pr[\text{PrivK}_{A,\Pi}^{eav}(n) = 1] \leq \frac{1}{2} + \text{negl}(n)$$

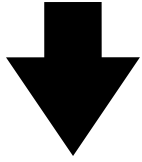
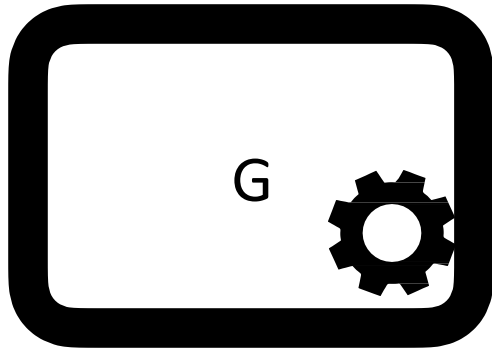
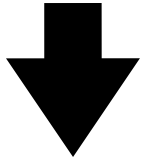
Stream Cipher

$$G: \{0, 1\}^s \rightarrow \{0, 1\}^n$$



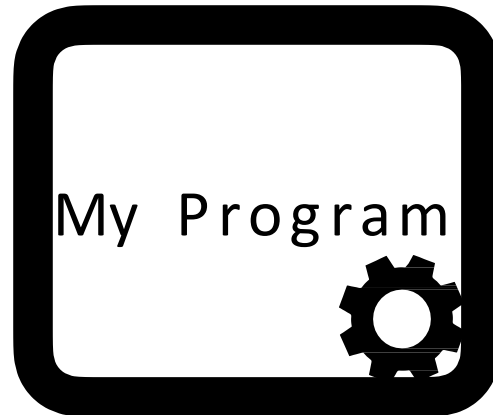
- Efficiently computable by a deterministic algorithm
- $s \ll n$

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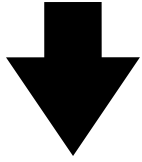
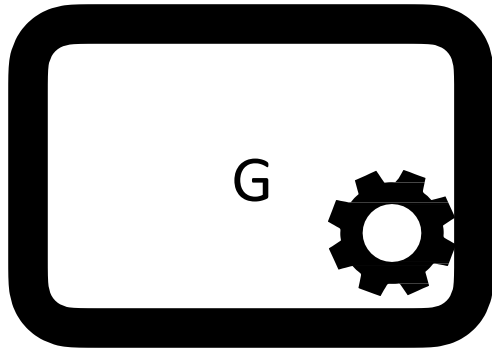
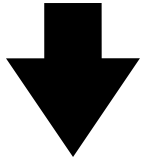
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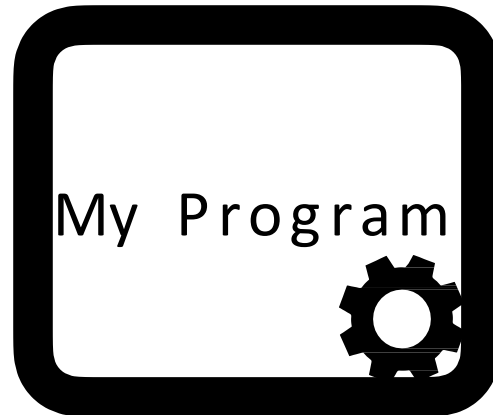
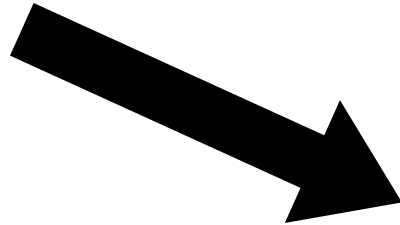


Slide credit to David Heath

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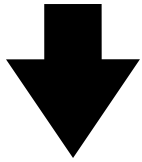
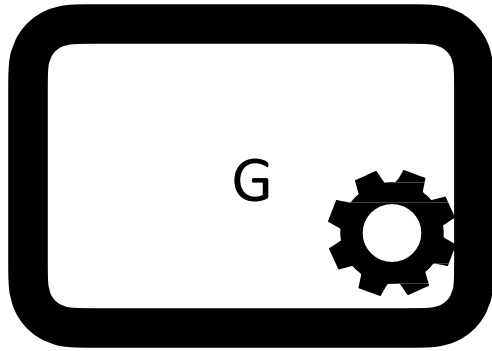
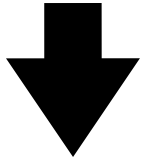
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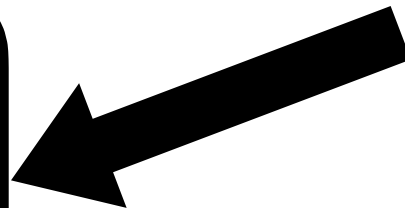
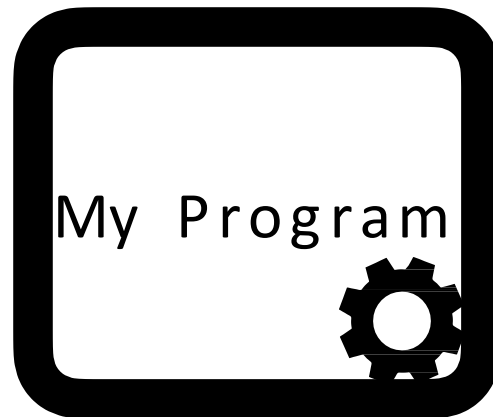


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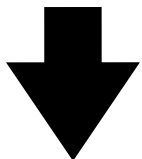
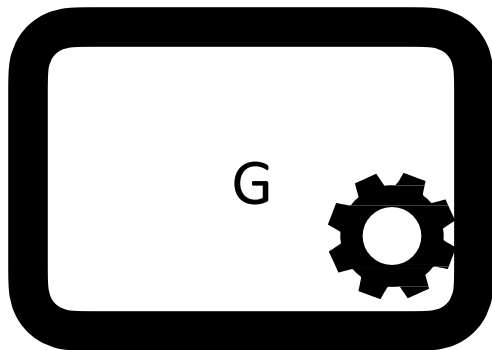
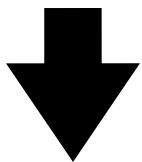


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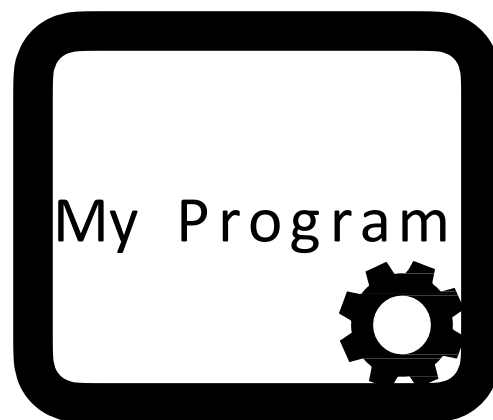
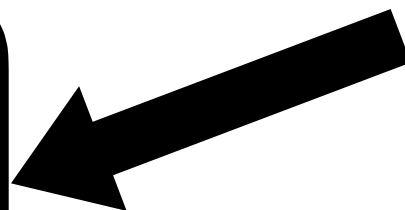
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**G is a PRG if *no* program can
reliably win this game**

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Definition

A PRG is a function $G : \{0,1\}^n \rightarrow \{0,1\}^{n+s}$

Pseudorandomness: for any PPT algorithm D , there is a negligible function negl such that

$$|\Pr[D(G(k)) = 1] - \Pr[D(r) = 1]| \leq \text{negl}(n)$$

For uniformly random $k \in \{0,1\}^n$ and $r \in \{0,1\}^{n+s}$

Security is defined as **observational equivalence** between protocol and its ideal functionality

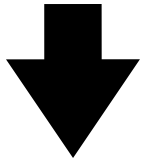
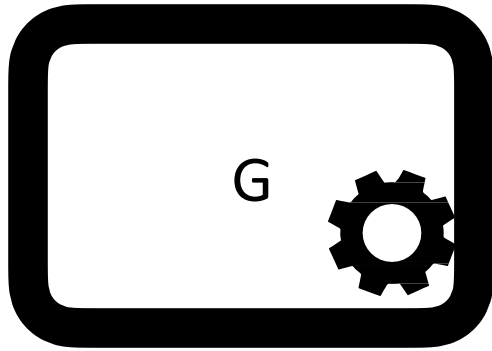
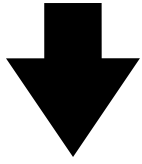
Example

- $G : \{0,1\}^n \rightarrow \{0,1\}^{n+1}$ by computing the XOR of the first n bits
- D : if $r_{n+1} = \bigoplus_{i=1}^n r_i$, output 1, otherwise, output 0.

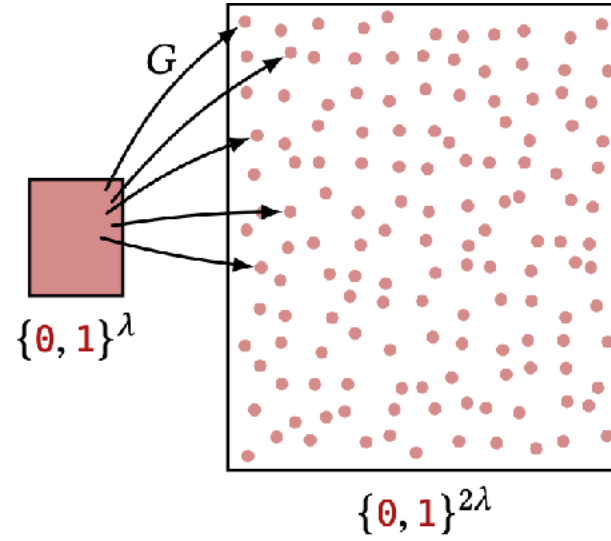
Brute-force search

- Distinguisher:
 1. precompute $G(s)$ for all possible $s \in \{0,1\}^n$
 2. If the string is in the table, output 1, otherwise output 0.
- What's $\Pr[D(G(k)) = 1]$?
- What's $\Pr[D(r) = 1]$ if $G : \{0,1\}^n \rightarrow \{0,1\}^{n+1}$?
- What's $\Pr[D(r) = 1]$ if $G : \{0,1\}^n \rightarrow \{0,1\}^{2n}$?

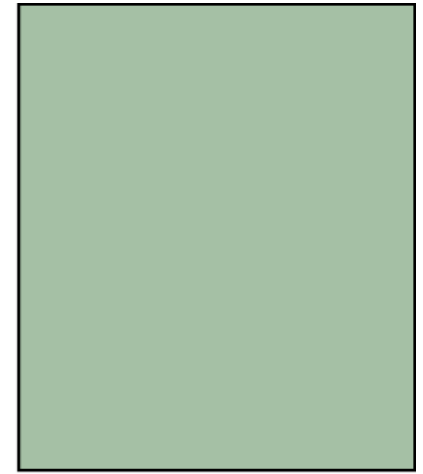
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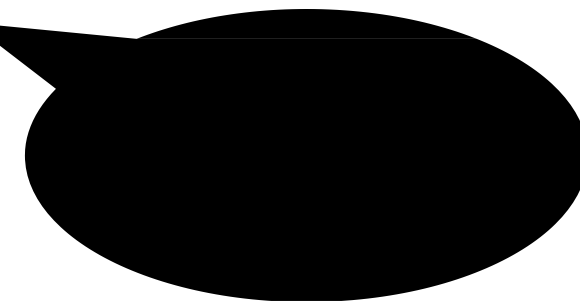
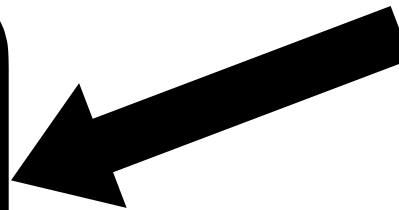
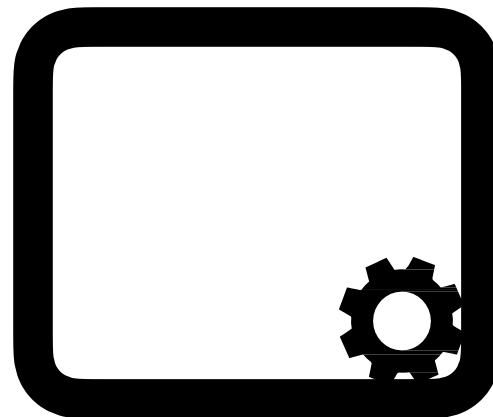
pseudorandom distribution



$\{0, 1\}^{2\lambda}$

uniform distribution

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EAV-security from PRG

Construction:

- $\text{Gen}(1^n)$: choose uniform $k \in \{0,1\}^n$ and output it as the key
- $\text{Enc}_k(m)$: on input a message $m \in \{0,1\}^{n+s}$, compute

$$c = G(k) \oplus m$$

- $\text{Dec}_k(c)$: on input a ciphertext $c \in \{0,1\}^{n+s}$, compute

$$m = G(k) \oplus c$$

EAV-security from PRG

- Theorem: if G is a PRG, then the construction is EAV secure.
 - Security from **observational equivalence** between protocol and its ideal functionality.
 - Pseudo one-time pad
- Proof.

Existence of PRG

One-way Permutation $\rightarrow$ PRG

- One-way permutation: for every poly-time A and $\text{negl}(n)$
$$\Pr[A(f(x)) = x] \leq \text{negl}(n)$$
- One-way permutation implies hardcore predicate $b: \{0,1\}^n \rightarrow \{0,1\}$: for every poly-time A and $\text{negl}(n)$

$$\Pr[A(f(x)) = b(x)] \leq \text{negl}(n) + \frac{1}{2}$$

$$\text{PRG } G: \{0, 1\}^n \rightarrow \{0, 1\}^{\{n+1\}} \quad \text{PRG}(x) = f(x) || b(x)$$

Proof of PRG security from OWP

- One-way permutation: for every poly-time A and $\text{negl}(n)$
$$\Pr[A(f(x)) = x] \leq \text{negl}(n)$$
- One-way permutation implies hardcore predicate $b: \{0,1\}^n \rightarrow \{0,1\}$:
for every poly-time A and $\text{negl}(n)$

$$\Pr[A(f(x)) = b(x)] \leq \text{negl}(n) + \frac{1}{2}$$

$$\text{PRG } G: \{0, 1\}^n \rightarrow \{0, 1\}^{n+1} \quad \text{PRG}(x) = f(x) || b(x)$$

Proof. Construct A against hardcore predicate using the distinguisher